Animation Pipeline

Author: Abdelkrim Mebarki

Master's Program in Computer Science – Artificial Intelligence & Applications
Department of Computer Science – Faculty of Mathematics and Computer Science
University of Science and Technology of Oran
December 26, 2025

In this context, it is essential to understand *how* animations are created. Each animation technique, whether traditional or digital, defines its creative process, tools, and workflow. Knowing these techniques helps students grasp the diversity of the artistic approaches and the technical requirements that shape every particular project. By exploring the different ways perceived motion can be produced — from the hand-drawn frames to the real-time 3D simulations — learners develop a complete sense of what's possible, what resources are needed, and which given methods best suit a given multimedia application.

Understanding the "**techniques**" provides the foundation for understanding the "**process**".

1. Global Classification of Animation Techniques

1.1. Traditional & Manual Animation Techniques

These techniques rely on the handcrafted artistic processes and include:

❖ Cut-Out Animation

Flat characters or objects cut from paper or photos are moved frame by frame under a camera.

❖ Chinese Shadow (Silhouette Animation)

Uses backlighting to project moving silhouettes. This is inspired by traditional shadow theatre.

❖ Painting, Pastel, and Sand Animation

Created/modified on glass, paper or sand, frame by frame, producing fluid, textural motion.

❖ Drawing on Film (Direct Animation)

Artists draw, scratch, or paint directly onto film strips, creating abstract or experimental effects.

❖ Cartoon Animation (Cel Animation)

Classic hand-drawn — Scenes are drawn on transparent sheets photographed frame by frame.

❖ Rotoscoping

Animators trace over live-action footage to reproduce realistic movement.

> 🎬 **Rotoscoping** is an **animation technique** where animators **trace over live-action footage frame by frame** to reproducing a realistic movement. This is often used to capture natural human motion, refine the complex gestures, or integrate the animated elements into real filmed scenes.
>
> For example –
>
> ➔ In traditional animation, artists projected the filmed images onto a light table and drew over them (Disney used this in his work *Snow White*).
>
> ➔ In modern digital workflows, software like *After Effects* or *Blender* allows digital rotoscoping to isolate subjects or recreate the motion paths.

❖ Collage Animation

Animated compositions created from the magazine cut-outs, any textures or photos.

❖ Paint-on-Glass Animation

Wet paints are moved and photographed frame by frame on glass surface for smooth transitions.

1.2. Stop-Motion and Physical Animation

Animation created by photographing real, physical objects frame by frame.

❖ Puppets and Figurines

Three-dimensional puppets are moved incrementally and photographed to create motion.

Available on zenodo – DOI : 10.5281/zenodo.18060487

❖ Clay Animation (Claymation)

This technique uses malleable clay models for flexible and expressive motion.

> 🧱 **Clay Animation (Claymation)** is a **stop-motion animation technique** using **malleable clay figures or models** to create motion. Each character or object is sculpted from **plasticine or modeling clay**; it is placed on a miniature set and **photographed frame by frame**. Between each two shots, the animator slightly moves the clay model — and when all frames are played in sequence, the figures in the scene appear to move naturally. This is widely known for its **expressive, handmade aesthetic** and its ability to give life to tactile, textured characters.
>
> 🎞️ **Famous examples –**
>
> ➔ *Wallace and Gromit* (Aardman Animations)
>
> ➔ *Chicken Run*
>
> ➔ *The Nightmare Before Christmas* (mixing clay and puppets)

❖ Pixilation

In this technique, we use live human actors as frame-by-frame “puppets,” for surreal movements.

> 🕴️ **Pixilation** (*pixillation*) is a **stop-motion animation technique** where **real humans** are used **as if they were stop-motion puppets**. The actors pose, move slightly between each two shotd, and are photographed frame by frame for a jerky, surreal, or magical effect when played.
>
> 🎬 **Example –** *Neighbours* (1952) by Norman McLaren (a classic example).

❖ Object Animation

Everyday objects are animated frame by frame to create illusion of life.

❖ Time-Lapse Animation

Captures slow processes (like a flower blooming) over long periods, then speeds them up.

> This is a **filming technique** (It is not a stop-motion, but similar in spirit) where **a camera captures images at long time intervals** — For example, one frame every few seconds, minutes, or even hours — and then plays them back at normal speed.
>
> ⌛ The result is a slow process like a flower blooming, a cloud moving, or a building being constructed, appear to happen **very fast**, almost magically.

1.3. Digital & Computer-Based Animation

This technique family relies on digital tools and software to create or manipulate movement.

❖ 2D Digital Animation

Vector or bitmap graphics animated digitally (Using Adobe Animate, Toon Boom or Synfig).

❖ 3D Animation

Objects modeled in 3D and animated virtually using keyframes rigging, and rendering.

❖ Motion Graphics

Animated shapes, icons or text —Synchronized with sound for multimedia interfaces or advertising.

❖ Kinetic Typography

Animated text conveying emotion, rhythm, or emphasis in multimedia presentations.

❖ Morphing

Smooth homeomorphic transformation from one image or shape into another.

❖ Procedural Animation

Motion generated algorithmically or through physics simulations.

❖ Rotoscoped Digital Animation

Modern digital tracing of live-action motion for realistic-like motion (For stylized games and films).

1.4. Interactive & Multimedia-Specific Animation

These animations are designed for **interactive real-time environments**.

❖ Scripted (Code-Based) Animation

Generated dynamically by programming (JavaScript, Unity, Unreal, Blender scripts).

❖ Data-Driven Animation

Movement or visuals change according to data inputs (sound, sensors, user input).

❖ UI/UX Micro-Animations

Small interface animations guiding users or giving feedback (buttons, transitions, hover effects).

❖ AR/VR Animations

Animated 3D elements integrated into immersive or mixed reality environments.

❖ Real-Time Engine Animation

Uses game engines for live rendering and interactivity (Unreal, Unity, Godot).

1.5. Hybrid & Experimental Techniques

Here, we combine multiple methods or explore unconventional materials.

❖ Mixed-Media Animation

Blends live-action, drawing, photography, and CGI in a single multimedia piece.

❖ Experimental & Abstract Animation

Focuses on rhythm, motion, and color rather than narrative (optical or musical visualizations).

❖ Augmented Performance Animation

Animation linked to physical gestures, sensors, or live motion capture.

2. The Animation Production Pipeline

The **general production pipeline** for creating **2D or 3D animations** follows always three major phases: **Pre-production**, **Production**, and **Post-production**. Each stage is essential for ensuring both artistic quality and technical consistency of the final animated scene sequence.

2.1. Pre-Production Stage

This phase defines the creative and the conceptual framework of the final animation involving planning, visual development, and narrative construction before any motion is really produced:

❖ Scenario Specification (Story Development)

The story or multimedia content provides the foundation of the animation. This stage defines:

- ➔ **Synopsis (Overview) –** That describes the nature of the plot, the setting (time & space), the main events, and all the characters involved.
- ➔ **Sequencer –** It breaks down the story scene by scene to guide visual and temporal organization.

❖ Character Design (Workbook Development)

It gives visual identity and personality to the protagonists and secondary figures. This stage defines:

- ➔ **Model Sheet (Character Board) –** Depicts each character's physical traits, attitudes, and typical poses through progressively: Skeletal structure → Simplified form → Finalized version.
- ➔ **Turnarounds –** A set of standard views (Front, Profile, Back, Three-Quarter Face, and Three-Quarter Back) used to maintain consistency in drawing and modeling over the whole process.
- ➔ **Facial Expression Board –** Illustrates the full range of emotions and expressions.
- ➔ **Lip-Sync and Mouth Chart –** Maps visemes (visual equivalents of phonemes) to ensure accurate synchronization between speech and animation.

❖ Storyboard Specification

The storyboard visually translates the content scenario into a sequence of illustrated frames (thumbnails) that serves as a blueprint for the following production process. Each panel indicates:

- ➔ The action & its temporal rhythm.
- ➔ Camera angles & framing.
- ➔ Locations, atmosphere, & lighting intentions.
- ➔ Character movements, dialogues, & sound effects.
- ➔ Emotional tone conveyed by the scene.

2.2. Production Stage

This main phase transforms the conceptual material into animated sequences through the visual creation and the motion synthesis.

❖ Setting Up Decorative Elements

Here, we design and organize **sets, backgrounds, and environments** according to each shot or layer. Fixed entities and decorative components are established to provide spatial coherence.

❖ Animation Process (25–30 frames per second)

The animation involves generating motion through the successive produced frames:

➔ **Key Poses Generation –** The main animators create essential poses defining the movement.
➔ **Refinement of Key Poses –** The assistant animators clean and finalize the key drawings.
➔ **Interpolation (Inbetweening) –** Inbetweeners produce the intermediate frames (tweens) that ensure smooth transitions between the key poses.

❖ Coloring

Consist of applying color to characters and backgrounds according to the visual identity defined during pre-production. This may involve digital painting, texture mapping, or shading processes.

2.3. III. Post-Production Stage

This final phase integrates audiovisual components and ensures the coherence of the animation.

➔ **Shooting (Compositing) –** Assembly of the visual layers (characters, backgrounds, effects) into a final animated sequence.
➔ **Sound Integration –** Recording, synchronization, and mixing of dialogues, sound effects, and music to support the content visual rhythm.
➔ **Editing –** Adjustment in timing, transitions, and visual effects achieving narrative/aesthetic unity.

Stage	Main Focus	Key Outputs
Pre-Production	Concept and planning	Script, character sheets, storyboard
Production	Animation creation	Animated sequences, colored assets
Post-Production	Integration and finalization	Final video, sound mix, editing

3. Organization, Cost, and Creative Constraints

The production of an animation (traditional, digital, or 3D) is a high **complex and resource-intensive process**. Beyond its underlying artistic dimension, it mobilizes significant **human, temporal, and financial resources**, requiring a correct coordination of multiple specialized roles and technologies. The animation pipeline, structured into pre-production, production, and post-production, functions as a **creative system** and as an **industrial workflow** at the same time.

3.1. A Costly and Time-Consuming Process

The animation production is inherently **expensive in time and labor**. Each second of animation may require **dozens of drawings or rendered frames**, detailed scene compositions, and synchronized sound. The duration and quality of the final product depend directly on:

➔ The **number of involved professionals** (artists, animators, sound engineers, editors),
➔ The **complexity of motion and effects**,
➔ The **level of realism or stylistic refinement** pursued.

In traditional and digital 2D animation, the process remains largely **manual**, despite a partial use of automation tools. In 3D production, while rendering and rigging pipelines reduce some repetition, **setup, modeling, and rendering times** still represent a major cost factor.

3.2. The Human Structure: Key Animators, Assistants, and Inbetweeners

Within production phase, **animation team** is hierarchically organized around creative functions:

➔ **Key (Lead) Animators** – They define the principal poses, gestures, and emotional states — Key animators give rhythm and personality to the movement.
➔ **Assistant Animators** – They refine these poses, ensure continuity, and adjust the timing and the proportion between the shots.

➔ **Inbetweeners** – They create the intermediate frames (or "tweens") that ensure smooth motion and fluid transitions.

This hierarchical organization is essential to guarantee **precision, consistency, and efficiency** in large productions where multiple sequences are handled simultaneously by different teams. In digital workflows, the same structure often persists except of "inbetweening", that can be **partially automated** through interpolation algorithms.

3.3. Between Creation & Automation

The animation production is a project that lies at the intersection of **human creativity** and **technical automation**. At its core, it remains an **art of movement and expression**, requiring human imagination, aesthetic sensitivity, and mastery of drawing and modeling abilities. The **creation and animation stages** depend on the artist's ability to translate emotions, rhythm, and story into visual motion.

Yet, as the process evolves from **storyboard to animation**, the automation progressively intervenes. The storyboard acts as a **blueprint** that guides camera angles, timing, and the shot composition. The software tools automate repetitive aspects: Interpolation, coloring, or layer compositing. But theses tools never replace the **creative decision-making** that drives the expressive and the narrative coherence.

3.4. The Logic of Shot-by-Shot Production and Layer Reusability

Animation progresses **shot by shot** (or scene by scene). This modular organization enables a **better management of complexity**, where each shot can be designed, animated, and rendered independently.

A key optimization technique is the **reusability of layers** — backgrounds, character rigs, and effects can be reused across multiple scenes in order to save time and ensure visual consistency. In 2D animation, for instance, the fixed backgrounds are often separated from the moving characters to allow only specific layers to change from one frame to another. In 3D animation, this principle translates into **asset libraries** and **modular scene components**.

However, this efficiency has some limits: reusability must not compromise the **organic flow** and the emotional authenticity. Excessive reuse can lead to mechanical and repetitive movement.

3.5. The Constraint of Non-Modifiable Visual Support

A particular difficulty of traditional and raster-based digital animation lies in its **non-revisable definition** once frames are finalized. Unlike vector graphics or 3D models, which can be modified dynamically, traditional drawings and painted frames constitute **non-modifiable visual supports**: They have *not editable source once rendered*.

This constraint means that any change in timing, background, or color after the fact may require **recreating or re-rendering** the entire sequences, which can be both time-consuming and costly. That's why, **rigorous planning** during pre-production (storyboarding, character design, and scene breakdown) is very essential to minimize later revisions.

4. Conclusion

The animation production pipeline is not only a sequence of technical operations; it is a **delicate balance between the human artistic intention, the team organization, and the technological constraint**. Its efficiency depends as much on the **clarity of human creative vision** as on the **precision of the production management**. Understanding its cost, roles, and limits helps future creators appreciate that every animated image is the result of both **imagination and discipline**, **freedom and structure**.